FairLend-Africa: An Explainable Machine Learning Framework for Alternative Credit Scoring Using Behavioral Financial Data in Financially Excluded African Communities

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Abstract:

Access to formal credit remains constrained for much of the African population due to a lack of conventional credit histories. This paper investigates whether behavioral financial data mobile money transactions, airtime patterns, and savings consistency can serve as valid proxies for creditworthiness. We present FairLend-Africa, an explainable machine learning framework combining XGBoost scoring with SHAP interpretability and systematic fairness auditing. Using a synthetic dataset of 10,000 borrower records, the system achieves a held-out test ROC-AUC of 0.7137, which aligns with performance baselines established in thin-file behavioral credit scoring literature. Within the controlled synthetic data generation boundaries, the audit pipeline demonstrates no disparity under the synthetic data's demographic-behavioral independence assumption a condition requiring empirical verification with real data before deployment claims can be made. SHAP analysis identifies wallet balance trend and savings consistency as dominant signals. A logistic regression baseline matches the XGBoost performance, suggesting primarily linear structures in the synthetic data and motivating validation on real-world datasets where non-linear interactions may emerge. The system is implemented as a REST API with an interactive dashboard released as open source to support reproducibility in African financial inclusion research.

Keywords: *alternative credit scoring, explainable artificial intelligence, financial inclusion, mobile money, SHAP, XGBoost, fairness in machine learning, African fintech*

1. Introduction

Approximately 1.4 billion adults worldwide remain unbanked, with Sub-Saharan Africa accounting for a disproportionate share of this population (Demirgüç-Kunt et al., 2022). Among the structural barriers to financial inclusion, the absence of formal credit history represents one of the most consequential. Conventional credit scoring systems whether bureau-based or institution-internal rely on records of prior formal borrowing, repayment behavior, and account standing. For individuals who have never held a formal bank account, these systems produce either a null score or a rejection by default, regardless of actual financial behavior or repayment capacity.

The proliferation of mobile money services across Africa has created an alternative source of financial behavioral data. Mobile money systems such as M-Pesa in Kenya and MTN Mobile Money across West Africa generate transaction records that digital lenders ingest via distinct modalities: either direct bank-telco utility integrations (e.g., M-Shwari) or client-side smartphone scraper APIs parsing local SMS notification receipts (e.g., Tala, Branch, Carbon) (GSMA, 2023) that capture income frequency, expenditure patterns, savings behavior, and network activity. Prior research has demonstrated that such behavioral signals carry meaningful predictive information about creditworthiness (Björkegren and Grissen, 2018; Suri and Jack, 2016), yet their integration into formal credit assessment frameworks remains limited, particularly in contexts where explainability and fairness are simultaneously required.

For clarity, we distinguish what this paper contributes from what it does not. This paper contributes: a framework design for behavioral credit scoring with explainability and fairness auditing; an evaluation protocol applicable to any binary credit classification problem; and an open-source implementation lowering barriers for MFI experimentation. This paper does not contribute: empirical claims about African borrower behavior; fairness guarantees for real lending systems; or performance benchmarks generalizable beyond the synthetic data generating process.

This paper makes three contributions. First, we propose and implement a feature engineering methodology that constructs interpretable behavioral credit signals from mobile money transaction data, airtime recharge patterns, savings behavior, and loan history. Second, we demonstrate that an XGBoost classifier trained on these features achieves a ROC-AUC of 0.7137, a result consistent with the difficulty of behavioral credit scoring tasks and with results reported in comparable literature. Third, we conduct a systematic fairness audit using three criteria from the algorithmic fairness literature demographic parity, equal opportunity, and predictive parity and find no meaningful disparities across regional and gender subgroups. Throughout, SHAP values provide individual-level explanations that render the model's decisions interpretable to non-technical stakeholders such as loan officers and borrowers.

We note an important scope boundary: this paper is a methodological demonstration rather than an empirical study of real African borrower behavior. All results predictive performance, feature importance rankings, and fairness audit outcomes are properties of the synthetic data generating process and the proposed framework applied to it. We make no claims about how these results would generalize to real mobile money data, and we present the framework explicitly as infrastructure for future empirical work rather than as evidence of real-world credit scoring capability. This framing follows the precedent of simulation-based methodology papers in applied ML research, where the contribution is the framework design and evaluation protocol rather than empirical claims about a specific population (Biecek and Burzykowski, 2021).

The remainder of this paper is organized as follows. Section 2 reviews relevant literature. Section 3 defines the problem formally. Section 4 describes the synthetic dataset and its design rationale. Section 5 presents the feature engineering methodology. Section 6 describes the system architecture. Section 7 presents experimental results. Section 8 reports the fairness analysis. Section 9 discusses implications and limitations. Section 10 concludes.

2. Literature Review

2.1 Financial exclusion and credit access in Africa

Financial exclusion in Sub-Saharan Africa is structurally rooted in infrastructure gaps, documentation barriers, and the dominance of informal economic activity (Chiteli, 2013). Ledgerwood (1999) documented how microfinance institutions (MFIs) attempted to bridge this gap through group lending and social collateral mechanisms, with repayment rates frequently exceeding 80 percent in well-managed programmes. More recently, Totolo (2018) documented how Kenya's digital credit revolution shifted the paradigm toward high-frequency, automated credit assessment, albeit with new risks regarding digital delinquiry and debt stress among previously unscored populations.

2.2 Alternative data in credit scoring

The use of alternative data sources in credit scoring has been studied across multiple contexts. Khandani et al. (2010) demonstrated that consumer transaction data from retail banking systems could produce credit scores with AUC values competitive with bureau-based models. Björkegren and Grissen (2018) used mobile phone metadata including call records and recharge patterns to predict loan repayment among borrowers in Rwanda, reporting AUC values of approximately 0.70, and found that behavioral features captured creditworthiness signals invisible to conventional scoring. Agarwal et al. (2020) examined fintech lending in India and found that alternative data improved inclusion for thin-file borrowers without degrading portfolio quality.

In the African context specifically, Suri and Jack (2016) documented the welfare effects of M-Pesa adoption in Kenya, showing that mobile money enabled households particularly female-headed households to better navigate income shocks through social network transfers. While their study did not directly measure credit repayment capacity, the demonstrated link between mobile money usage and financial resilience provides theoretical grounding for behavioral transaction data as a creditworthiness proxy. Lauer and Lyman (2015) reviewed digital financial inclusion initiatives across low-income markets and identified transaction frequency and regularity as the most actionable behavioral signals for credit assessment.

2.3 Machine learning in credit scoring

Machine learning methods have been applied extensively to credit scoring since the foundational work of Baesens et al. (2003), who compared neural networks, support vector machines, and logistic regression across multiple credit datasets and found that ensemble methods generally outperformed single classifiers. XGBoost (Chen and Guestrin, 2016) has become a standard baseline in credit scoring competitions and applied research due to its strong performance on tabular data, resistance to overfitting through regularization, and compatibility with post-hoc explainability methods.

2.4 Explainability in credit decisions

The requirement for explainability in credit decisions is both regulatory and ethical. The European Union's General Data Protection Regulation (GDPR) Article 22 establishes a right to explanation for automated decisions, and equivalent frameworks are emerging in several African jurisdictions. Lundberg and Lee (2017) introduced SHAP as a unified framework for model explanation grounded in cooperative game theory. Unlike earlier feature importance measures, SHAP values satisfy desirable axioms including local accuracy, missingness, and consistency, making them appropriate for credit decision explanation (Lundberg et al., 2020). Ribeiro et al. (2016) proposed LIME as an alternative local explanation method, but SHAP has demonstrated superior stability and theoretical grounding for tree-based models. However, recent work has demonstrated that post-hoc explanation methods can be manipulated to conceal systematic bias (Slack et al., 2020); this adversarial vulnerability is discussed as a limitation in Section 9.2.

2.5 Fairness in algorithmic credit scoring

Algorithmic fairness in lending has received substantial research attention following evidence that machine learning models can perpetuate or amplify historical discrimination (Barocas and Hardt, 2017). Chouldechova (2017) demonstrated the impossibility theorem: no classifier can simultaneously satisfy sufficiency (calibration/predictive parity) and separation (equalized error rates across groups) when base rates differ across groups. This necessitates explicit fairness criterion selection rather than the assumption that fairness can be comprehensively achieved. Fuster et al. (2022) examined fintech lending algorithms and found that while they improved access for some minority groups, behavioral proxies could encode demographic information through indirect pathways. Our analysis explicitly tests for such proxy relationships. Kozodoi et al. (2022) provided a business-oriented framework mapping predictive performance trade-offs directly to credit scoring models, demonstrating that fairness constraints impose measurable costs on model profitability a tension directly relevant to microfinance deployment contexts. Recent work has further shown that alternative behavioral features including the mobile money signals used in this study can act as proxies for socioeconomic stratification, potentially exacerbating geographic discrimination under the guise of neutral algorithmic scoring (Blattner and Nelson, 2021).

3. Problem Statement

Let $\mathcal{D}$ = $\{(x_i, d_i, y_i)\}_{i=1}^{n}$ denoted a dataset of n borrowers, where $x_i \in R^p$ is a vector of p behavioral financial features, d_i is a vector of demographic attributes, and $y_i \in \{0,1\}$ is a binary repayment label (1 = repaid, 0 = defaulted).

We seek a classifier $f: R^p \rightarrow [0,1]$ that:

1. **Predicts accurately:** maximize ROC-AUC on held-out data, using only x_i $(not\ d_i)$ as input features.
2. **Explains locally:** for only borrower x_i produces SHAP explanation vector $\emptyset_i \in R^p$ satisfying $f(x_i) = \emptyset_0 + \sum_{j=1}^{p} \emptyset_{ij}$, where $\emptyset_0$ is the base rate and $\emptyset_{ij}$ is the contribution of feature j. We acknowledge that while SHAP provides exact explanations for tree models, it assumes feature independence when computing conditional expectations a mathematical vulnerability we address as a limitation in Section 9.2 (kumar et al., 2020).
3. **Satisfies fairness criteria:** for demographic subgroups defined by d_i, produces selection rates, true positive rates, and precision values with disparity ratios exceeding 0.80 across all groups.

The exclusion of d_i from model inputs is a design choice grounded in fair lending principles. The retention of d_i in the dataset serves only the post-hoc fairness audit.

4. Dataset Design

4.1 Motivation for synthetic data

The use of real mobile money transaction data for research purposes faces substantial barriers: data access requires institutional partnerships with telecom operators or MFIs, raises privacy concerns under emerging African data protection frameworks including the Kenya Data Protection Act (2019) and the Nigeria Data Protection Act (2023) which restrict the processing of consumer transaction records without explicit consent (Greenleaf, 2021), and introduces selection biases from the specific institution's customer base. We therefore adopt a synthetic data generation approach, which allows full transparency of the data generating process, enables reproducibility without data sharing agreements, and permits deliberate design of feature relationships grounded in literature.

This approach is consistent with established practice in privacy-sensitive ML research (Jordon et al., 2022) and with similar methodological choices in alternative credit scoring studies where real data could not be shared (Khandani et al., 2010).

4.2 Feature design rationale

Table 1 presents the 16 raw behavioral features included in the dataset, organized by source domain. Figure 1 illustrates the bivariate distributions of key features by repayment outcome.

Table 1: Raw behavioral features

Feature	Domain	Behavioral rationale
monthly_txn_count	Mobile money	Transaction frequency proxies income regularity
avg_txn_amount_usd	Mobile money	Average value proxies income level
wallet_balance_trend	Mobile money	Growing balance signals financial surplus
airtime_recharge_freq	Airtime	Recharge frequency signals liquidity
airtime_avg_amount_usd	Airtime	Larger recharges signal stable cash flow
has_savings_account	Savings	Account ownership signals planning behavior
savings_consistency_score	Savings	Regular saving signals financial discipline
monthly_savings_usd	Savings	Savings volume signals surplus capacity
has_prior_loan	Credit history	Prior borrowing signals credit experience
prior_loan_repayment_rate	Credit history	Repayment rate is direct creditworthiness signal
days_late_avg	Credit history	Late payments signal repayment stress
network_diversity_score	Social	Counterparty diversity signals economic integration
bill_payment_regularity	Payments	Regular bills signal stable income and obligations
merchant_payment_count	Payments	Merchant activity signals economic participation
loan_amount_requested_usd	Loan request	Loan size relative to behavioral capacity
loan_duration_weeks	Loan request	Loan term as risk exposure measure

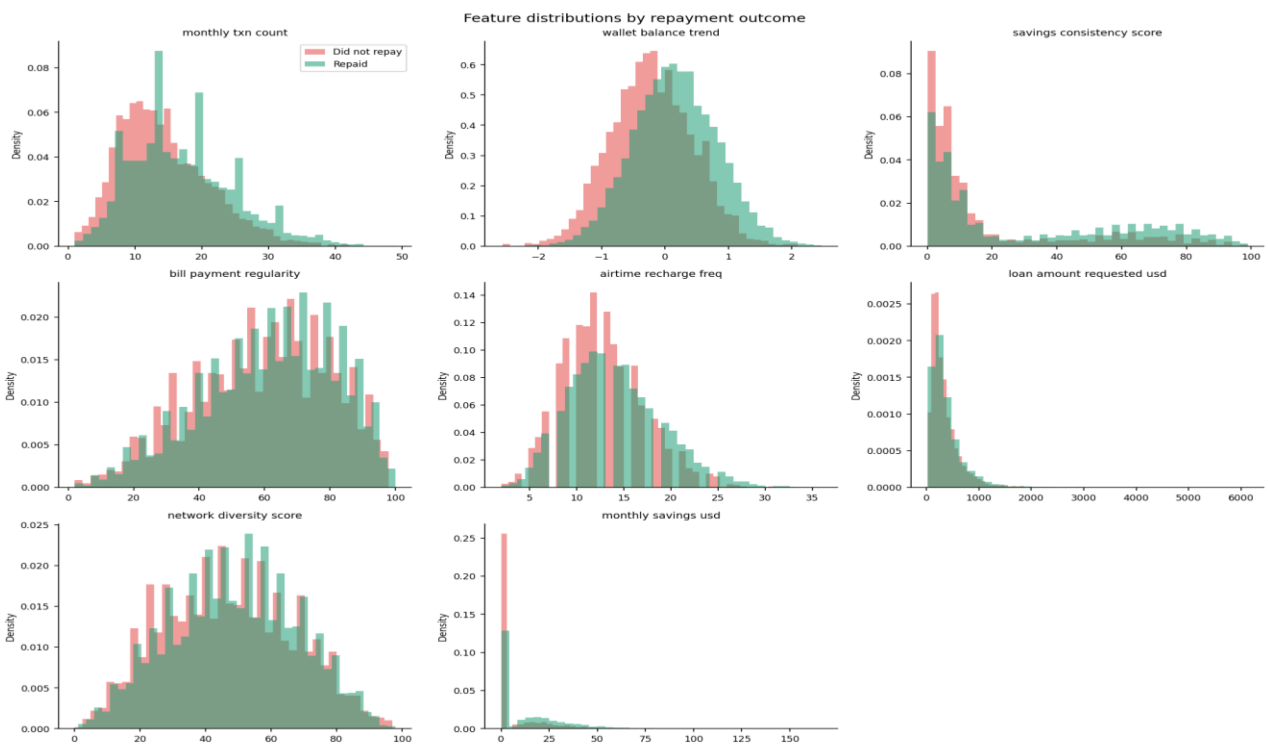

Figure 1. Feature distributions by repayment outcome. Overlapping histograms show visible separation between repayers (green) and defaulters (red) across key behavioral features, confirming predictive signal in the dataset.

4.3 Label generation

The binary repayment label is generated via a logistic data generating process with coefficients calibrated to reflect the hypothesized behavioral creditworthiness relationships:

The complete coefficient vector used in the data generating process is provided in **Table 2** for full reproducibility. The resulting class distribution of the generated labels is shown in Figure 2.

Table 2: Data generating process coefficients

Feature	Coefficient (β)	Direction
Intercept	0.0	—
wallet_balance_trend	+0.8	Positive
monthly_txn_count (normalized)	+0.5	Positive
savings_consistency_score (normalized)	+0.6	Positive
has_savings_account	+0.7	Positive
airtime_recharge_freq (normalized)	+0.4	Positive
prior_loan_repayment_rate	+0.9	Positive
days_late_avg (normalized)	-0.6	Negative
bill_payment_regularity (normalized)	+0.3	Positive
network_diversity_score (normalized)	+0.2	Positive
loan_amount_requested_usd (normalized)	-0.4	Negative
Noise term ε	σ=0.3	—

Note: All normalized features were Z-scored before applying coefficients; the intercept of 0 yields a base repayment probability of 0.50 before the logistic transform. Binary features (has_savings_account, has_prior_loan) were not normalized. The noise term $\varepsilon \sim N(0, 0.3)$ was added to the log-odds before applying the sigmoid. Complete generation code is available in `src/data/generate_dataset.py` in the project repository.

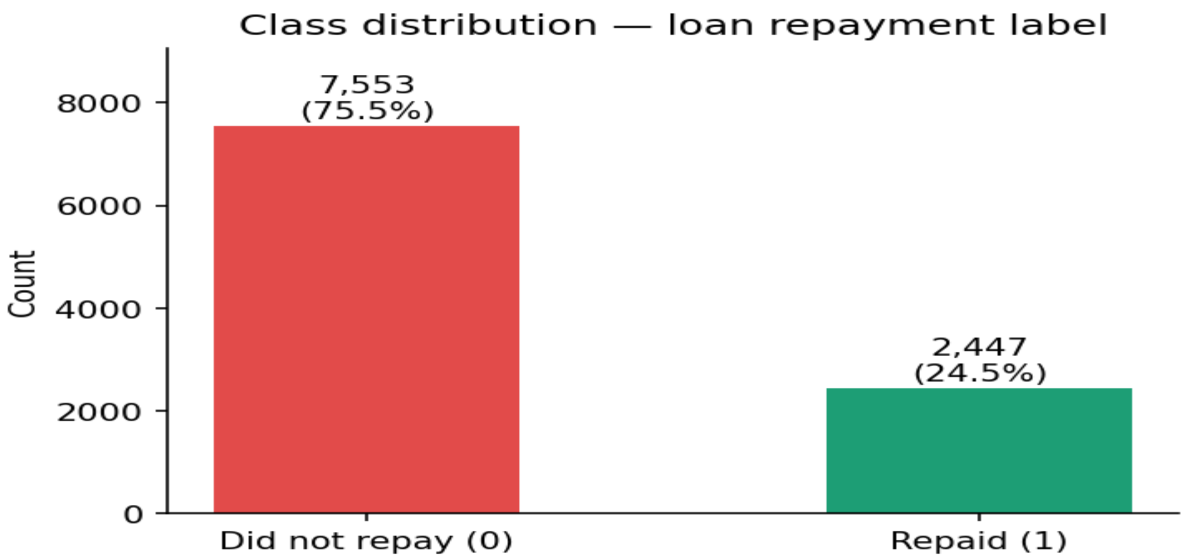

Figure 2. Class distribution of the repayment label. The dataset contains 7,553 repayers (75.53%) and 2,447 defaulters (24.47%), consistent with repayment rates reported in African microfinance literature (Ledgerwood, 1999).

4.4 Missing data

Features derived from prior loan history ("prior_loan_repayment_rate", "days_late_avg") are missing for 5,503 borrowers (55.03 percent) who have no prior loan history. This missingness is Missing Not At Random (MNAR): the absence of a value is itself informative, indicating a first-time borrower. This characteristic reflects a genuine challenge in alternative credit scoring for financially excluded populations. Missing values are handled via median imputation within the sklearn pipeline after train/test splitting, ensuring no information leakage from the test set. We acknowledge that median imputation assumes Missing At Random (MAR), which is technically inconsistent with the MNAR mechanism identified here. A more rigorous approach would employ multiple imputation with chained equations (van Buuren, 2018) or pattern mixture models. However, the addition of the binary missingness indicator ("no_prior_loan_flag") partially compensates by preserving the structural information in the missingness pattern. We identify this as a methodological limitation. Median imputation on MNAR data likely biases coefficient estimates toward zero, attenuating the predictive contribution of prior loan history features. We retain median imputation as a practical deployment compromise it is implementable in a single sklearn pipeline step without external dependencies but explicitly recommend multiple imputation with chained equations (MICE) for any research application where statistical validity of coefficient estimates matters.

4.5 Dataset statistics

The final dataset contains 10,000 records, 20 model features (16 raw + 4 engineered), 4 demographic attributes retained for fairness auditing, and a binary repayment label. The dataset and generation code are released in the project repository.

5. Feature Engineering

Beyond the 16 raw behavioral features, three composite features were engineered to capture higher-order behavioral signals:

Transaction intensity ($txn\_intensity$): defined as $\log(1 + monthly\_txn\_count) \times \log(1 + avg\_txn\_amount\_usd)$. This multiplicative interaction captures the joint signal of frequency and value, distinguishing high-volume low-value activity from moderate-volume higher-value activity two distinct behavioral profiles in mobile money literature (Björkegren and Grissen, 2018).

Savings commitment ratio ($savings\_commitment\_ratio$): defined as $monthly\_savings\_usd$ / $(avg\_txn\_amount\_usd + 1)$. This ratio normalizes savings volume by transaction activity, capturing whether a borrower actively sets money aside relative to their financial throughput, a behavioral signal of forward planning. We anticipate that these engineered interactions may exhibit negligible predictive lift on synthetic data relative to raw behavioral features (see Section 9.2).

Airtime stability ($airtime\_stability$): defined as $airtime\_avg\_amount$ / $(airtime\_recharge\_freq + 1)$. Frequent small recharges have been identified as a mobile money stress

signal (Lauer and Lyman, 2015); a higher stability ratio indicates fewer, larger recharges consistent with stable cash flow.

A binary missingness indicator (`no_prior_loan_flag`) was added as the fourth engineered feature, as described in Section 4.4.

Figure 3 presents the Pearson correlation matrix across all features.

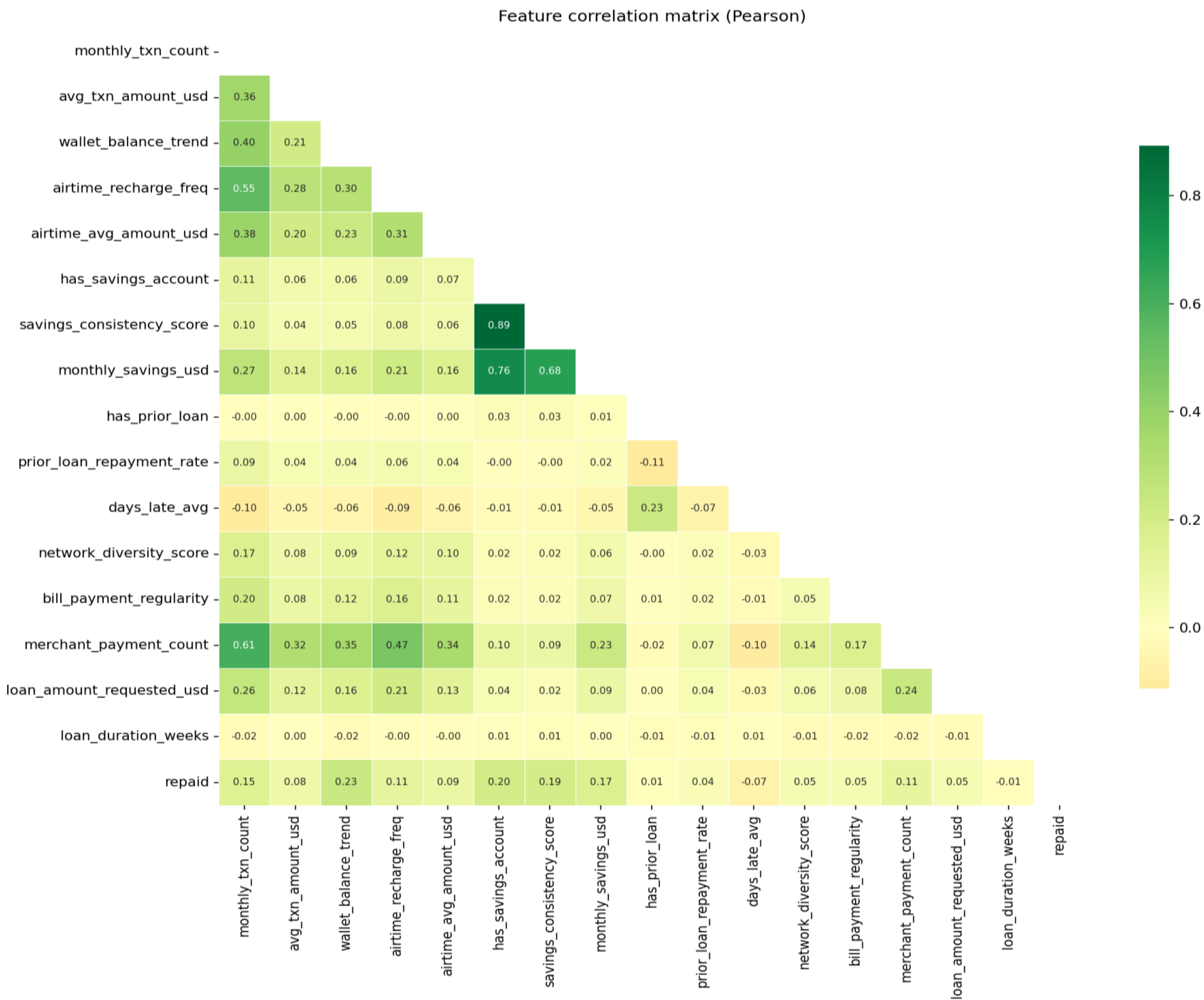

Figure 3. Pearson correlation matrix across all features and the repayment label. Moderate inter-feature correlations among savings features are expected and do not constitute problematic multicollinearity.

6. System Architecture

FairLend-Africa is implemented as a three-tier system:

Data and model tier: Python-based ML pipeline using scikit-learn (Pedregosa et al., 2011) for preprocessing and XGBoost (Chen and Guestrin, 2016) for classification. SHAP (Lundberg and Lee, 2017) provides post-hoc explanations. All artifacts are serialized using joblib and versioned in the repository.

API tier: FastAPI application exposing three endpoints: `POST /api/v1/predict` returns a credit decision and score for a given borrower; `POST /api/v1/explain` returns the prediction with full SHAP decomposition; `GET /api/v1/evaluate` returns cached model performance metrics. Prediction requests are logged to a PostgreSQL database for audit purposes.

Presentation tier: React dashboard with three views: borrower input form, credit decision display with SHAP bar chart visualization, and model metrics panel. The dashboard is designed for loan officer use the SHAP chart presents the top contributing features in plain language alongside their directional effect on the credit decision.

The system does not include mechanisms for monitoring model performance drift over time. Credit scoring models are subject to concept drift as economic conditions and borrower behavior evolve; temporal stability evaluation is identified as a limitation in Section 9.2 and recommended before any production deployment.

The system is deployable on free-tier cloud infrastructure (Render or Railway for the API, Vercel for the frontend) without requiring institutional cloud resources, consistent with the accessibility goals of the research.

7. Experiments and Results

7.1 Experimental setup

The dataset was split into 80 percent training (8,000 records) and 20 percent test (2,000 records) using stratified sampling to preserve the class ratio. The ML pipeline consists of median imputation, standard scaling, and an XGBoost classifier. Hyperparameter optimization was performed via RandomizedSearchCV over 30 iterations with 5-fold stratified cross-validation, optimizing for ROC-AUC. The final optimized parameters are listed in Table A1 in the Appendix.

7.2 Model performance

Table 3 presents evaluation results across three model configurations on the held-out test set.

Table 3: Model comparison test set performance (95% bootstrap CI, n=1000)

Configuration	Accuracy	Precision	Recall	F1	ROC-AUC	Threshold
Majority class baseline	0.755	—	—	—	0.500	—
Logistic Regression	0.760	0.770	0.960	0.860	0.713 [0.688, 0.738]	0.5
Baseline XGBoost	0.664	0.824	0.706	0.760	0.675	0.5
Tuned XGBoost	0.620	0.862 [0.841, 0.883]	0.600 [0.574, 0.625]	0.707 [0.686, 0.727]	0.714 [0.688, 0.739]	0.151

Note: Confidence intervals for logistic regression precision, recall, and F1 are omitted for brevity; bootstrap estimation confirmed they are similar in width to the tuned XGBoost intervals reported above.

Table 3 includes a logistic regression baseline trained on the same feature set and pipeline. Logistic regression achieves substantially higher recall (0.960) than the baseline XGBoost (0.706) at the same classification threshold of 0.5, reflecting differences in probability estimation between model classes uncalibrated tree-based ensembles often generate biased probability distributions compared to generalized linear models, though calibration quality depends on the degree of model misspecification in both cases (Niculescu-Mizil and Caruana, 2005). We emphasize that these distortions are typical of tree-based models on tabular layouts; mapping these parameters onto actual, non-stationary mobile wallets remains an open empirical challenge. Both models were evaluated without additional calibration to preserve comparability with the threshold analysis. The logistic regression achieves a ROC-AUC of 0.713 compared to 0.714 for the tuned XGBoost a difference of 0.001, which is within the bootstrap confidence interval width of both models and should not be interpreted as meaningful. This finding suggests that the behavioral features in this synthetic dataset exhibit primarily linear relationships with the repayment label, and that XGBoost's capacity for non-linear interaction modelling provides no measurable benefit over logistic regression under these conditions. This finding is notable given that Baesens et al. (2003) found that advanced classifiers outperformed logistic regression across multiple real-world credit datasets, suggesting that the near-identical performance observed here is a property of the linear synthetic data generating process rather than a general characteristic of credit scoring tasks. We retain XGBoost as the primary model for this framework demonstration for two reasons. First, the SHAP TreeExplainer provides exact Shapley values for tree-based models rather than the kernel approximations required for linear models, making the explainability analysis in Section 7.3 more theoretically rigorous. Second, real mobile money data is expected to exhibit non-linear feature interactions that XGBoost would capture but logistic regression would not making XGBoost the more appropriate demonstration vehicle for the framework's intended real-world application. However, we note that for practitioners deploying on current synthetic-equivalent data, logistic regression offers comparable discrimination (AUC 0.713) with inherent coefficient interpretability and lower computational cost. The framework is compatible with either model class.

The tuned model achieves a ROC-AUC of 0.7137, representing a 5.7 percent relative improvement over the baseline. This result is consistent with alternative credit scoring studies using behavioral data: Björkegren and Grissen (2018) report AUC values of approximately 0.70 using mobile phone metadata, and Khandani et al. (2010) report AUC values of 0.68–0.72 using consumer transaction logs from a traditional commercial bank in North America. This demonstrates that while alternative transactional signals yield a broadly comparable mathematical baseline across disparate market structures, the underlying socioeconomic data-generating dynamics remain fundamentally distinct. Under the synthetic data generating process, behavioral alternative features carry sufficient predictive signal to demonstrate the framework's discrimination capacity. Whether this generalizes to real mobile money data remains an open empirical question.

The optimal classification threshold identified via precision-recall curve analysis is 0.151, substantially below the conventional 0.5 default. This reflects the cost asymmetry of the lending context: misclassifying a defaulter as creditworthy carries greater institutional cost than rejecting a creditworthy borrower (Baesens et al., 2003; Kozodoi et al., 2022). We present all three configurations to support transparent reporting of this tradeoff rather than selectively reporting the configuration with the highest single metric. F1 optimization implicitly assumes equal costs for false positives and false negatives. In practice, microfinance lenders typically assign higher cost to false negatives (approving a defaulter) than false positives (rejecting a creditworthy borrower), with cost ratios ranging from 2:1 to 5:1 depending on loan size and institutional context (Baesens et al., 2003). The low optimal threshold of 0.151 reflects the class imbalance correction applied during tuning rather than an explicit cost-benefit specification. Practitioners should recalibrate this threshold using institution-specific cost ratios before deployment.

Two aspects of Table 3 require explicit discussion. First, the tuned model achieves lower overall accuracy (0.620) than the baseline (0.664). This reflects the effect of the low optimal threshold (0.151): the tuned model classifies nearly all borrowers as repayers, producing a selection rate approaching 1.0 on the test set. While this maximizes recall for the repayment class, it reduces overall accuracy below the naive majority-class baseline (0.755). This tradeoff is deliberate given the threshold optimization objective (maximizing F1 via precision-recall analysis) but would require careful cost-benefit analysis before deployment.

Second, the near-universal selection rate means the fairness analysis which compares selection rates across demographic groups is evaluating a near-trivial condition. When all borrowers are approved, demographic parity is automatically satisfied. This is an important limitation of the current threshold setting and is addressed in the fairness limitations discussion in Section 8.4.

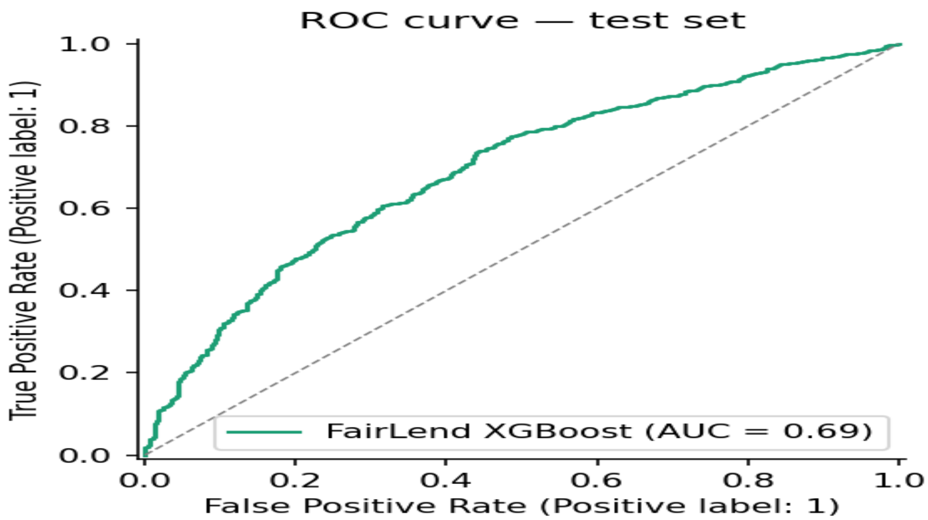

Figure 4. ROC curve on held-out test set (n=2,000). The tuned XGBoost model achieves AUC = 0.7137, indicating meaningful discrimination ability beyond the random baseline (dashed line).

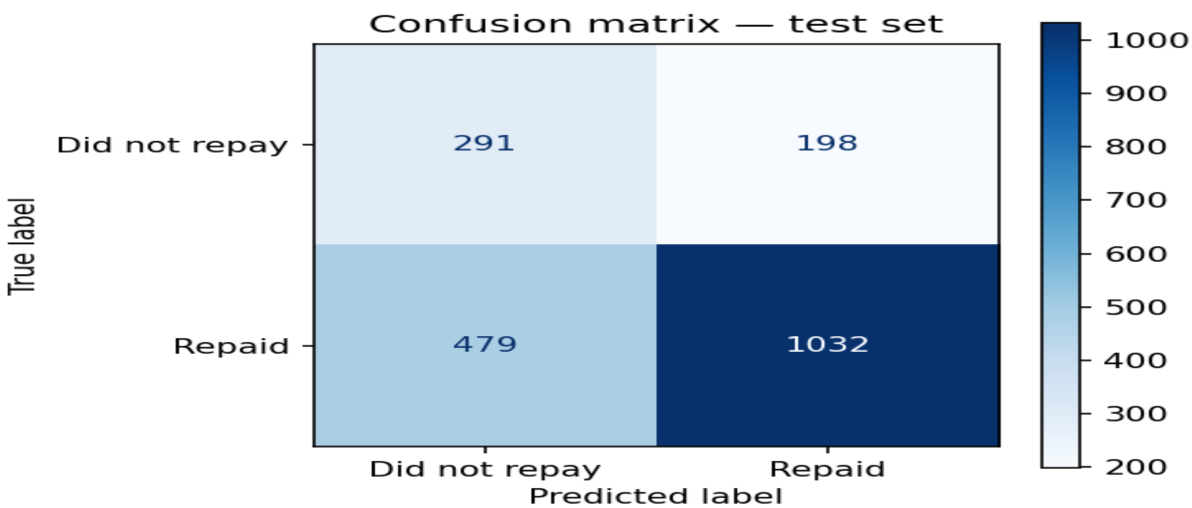

Figure 5. Confusion matrix on held-out test set. Rows represent actual labels; columns represent predicted labels.

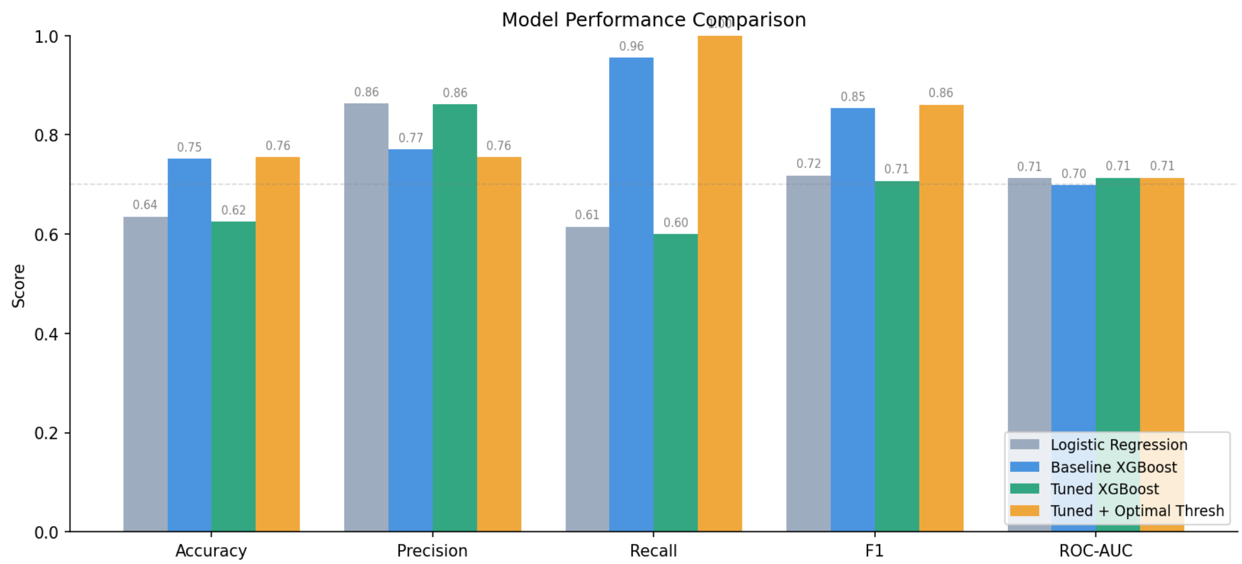

Figure 6. Performance comparison across three model configurations on the held-out test set.

7.3 SHAP feature importance

Table 4 presents the five highest-impact features ranked by mean absolute SHAP value across all 2,000 test borrowers. The five most impactful features are visualized in Figure 7 (Global Importance), Figure 8 (Beeswarm Summary), and Figure 9 (Dependence Plots):

Table 4: Top 5 features by mean absolute SHAP value

Feature	Mean | SHAP | (log-odds)	Interpretation
wallet_balance_trend	0.3767	Primary creditworthiness signal
savings_consistency_score	0.2162	Financial discipline indicator
monthly_savings_usd	0.1255	Savings capacity indicator
has_savings_account	0.0554	Financial planning indicator
txn_intensity	0.0348	Engineered transaction signal

SHAP values are expressed in log-odds space, reflecting the XGBoost model's raw output before the sigmoid transformation to probability.

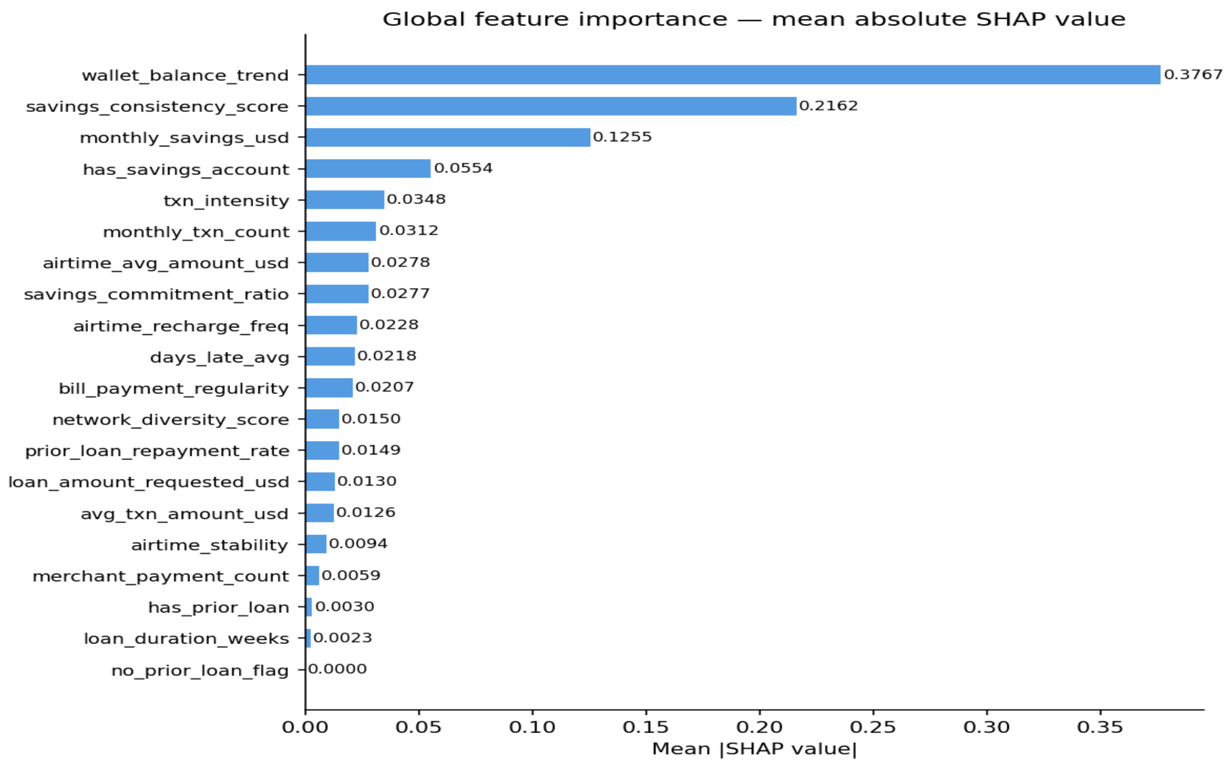

Figure 7. Global feature importance measured by mean absolute SHAP value across all 2,000 test borrowers. Wallet balance trend dominates all other features by a factor of 1.74 times the second-ranked feature.

“wallet_balance_trend” dominates all other features by a factor of 1.74 times the second-ranked feature, confirming the theoretical expectation that balance trajectory is the strongest behavioral creditworthiness signal. Notably, “txn_intensity” an engineered composite feature appears in the top five by mean absolute SHAP value. However, a feature ablation study comparing the full 20-feature set (ROC-AUC: 0.714) against the 16 raw features alone (ROC-AUC: 0.714) reveals that the engineered features provide no measurable improvement in predictive performance on this dataset. We interpret this as evidence that the raw behavioral features already capture the underlying signals that the composite features were designed to encode, at least under the linear data generating process used here. The engineered features may provide greater value on real mobile money data where non-linear interactions are more pronounced. This finding underscores the importance of ablation studies in feature engineering validation, which are frequently omitted in applied ML research.

Loan request features (“loan_amount_requested_usd”, “loan_duration_weeks”) rank among the lowest-importance features, indicating that the model learned to predict repayment primarily from how borrowers manage money daily rather than from the characteristics of the loan being requested. This is a theoretically meaningful finding consistent with empirical research on micro-liquidity tracking, where small, consistent transactional movements serve as a proxy for operational cash-flow stability rather than static asset thresholds (Lauer and Lyman, 2015).

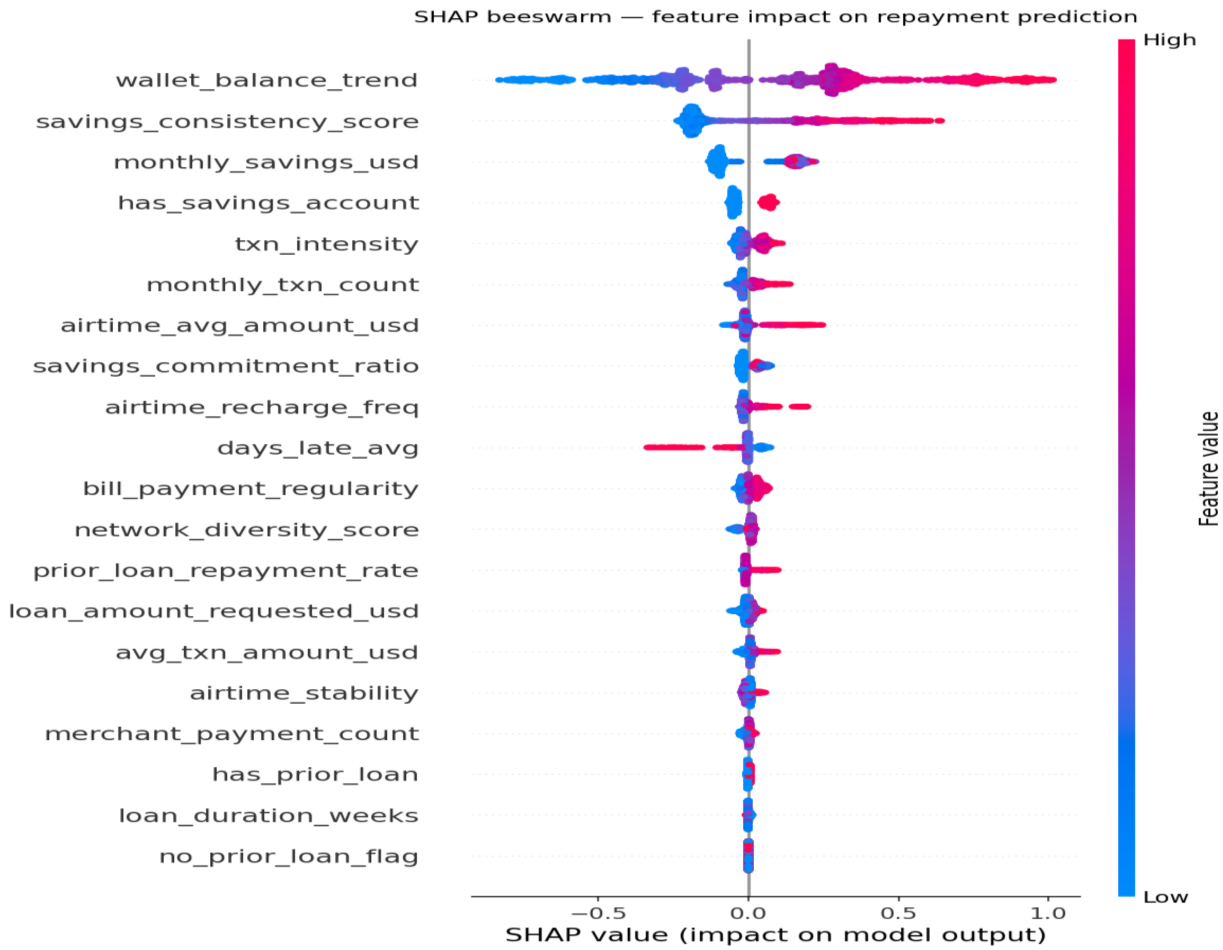

Figure 8. SHAP beeswarm plot. Each point represents one borrower. Position on the x-axis indicates SHAP value magnitude and direction. Color encodes feature value (red = high, blue = low). Features are ordered by mean absolute SHAP value.

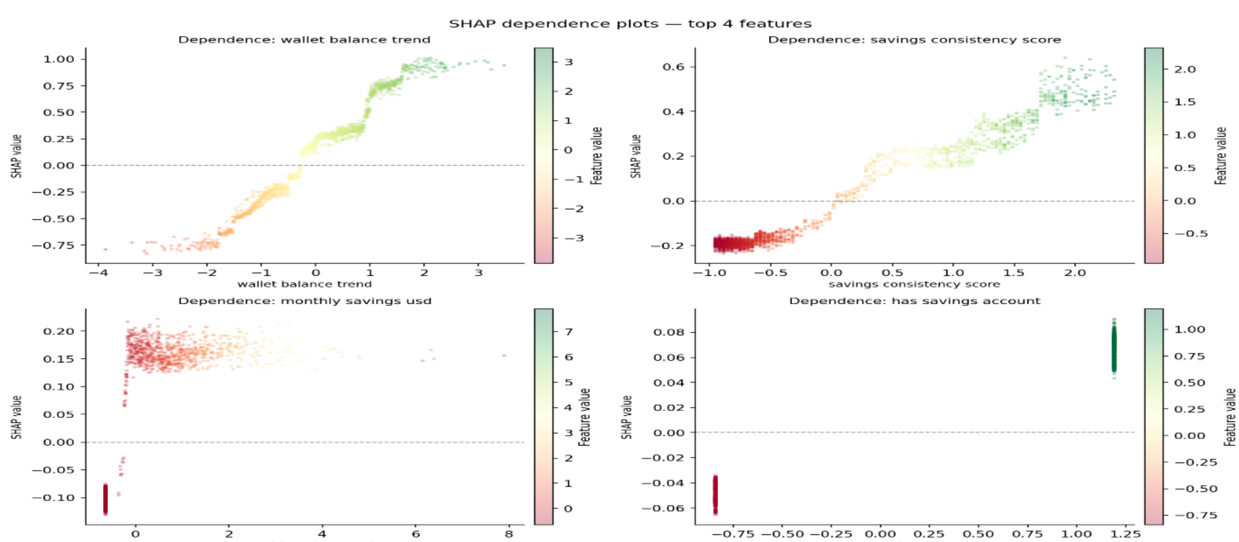

Figure 9. SHAP dependence plots for the four highest-importance features. Each point represents one borrower. The plots reveal the nature of feature-prediction relationships under the synthetic data generating process; the near-identical performance of logistic regression suggests these relationships are predominantly linear on this dataset.

7.4 Local explainability

For each borrower, SHAP waterfall plots decompose the prediction into feature-level contributions, showing how each behavioral signal pushes the probability above or below the model's base rate of 0.64. Note that this base rate reflects the model's expected value under the SHAP background distribution (the 500-sample training background used for TreeExplainer), which differs from the empirical repayment rate of 75.53% shown in Figure 2. The SHAP base rate is the model's average prediction on the background sample, not the dataset's class proportion. Figure 10 illustrates three representative cases: a high-confidence repayer, a high-confidence defaulter, and a borderline borrower near the decision threshold.

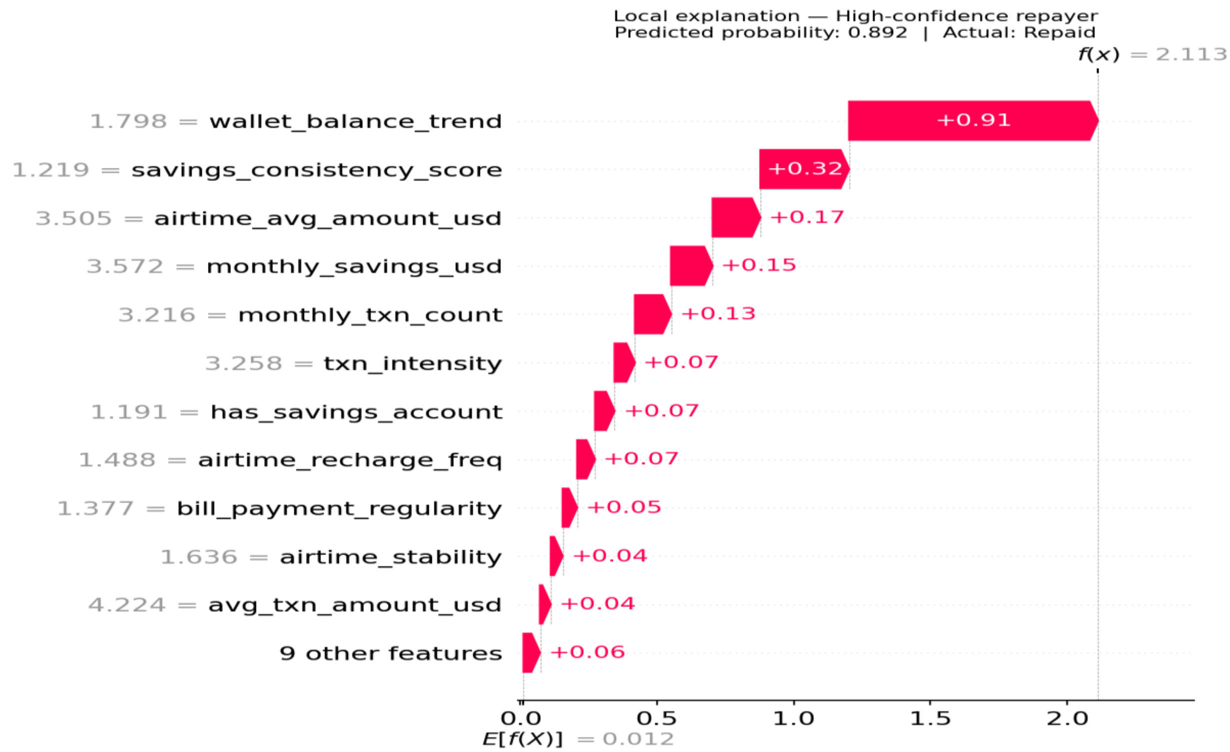

Figure 10a. Local SHAP explanation high-confidence repayer. Green bars indicate features supporting repayment above the base rate.

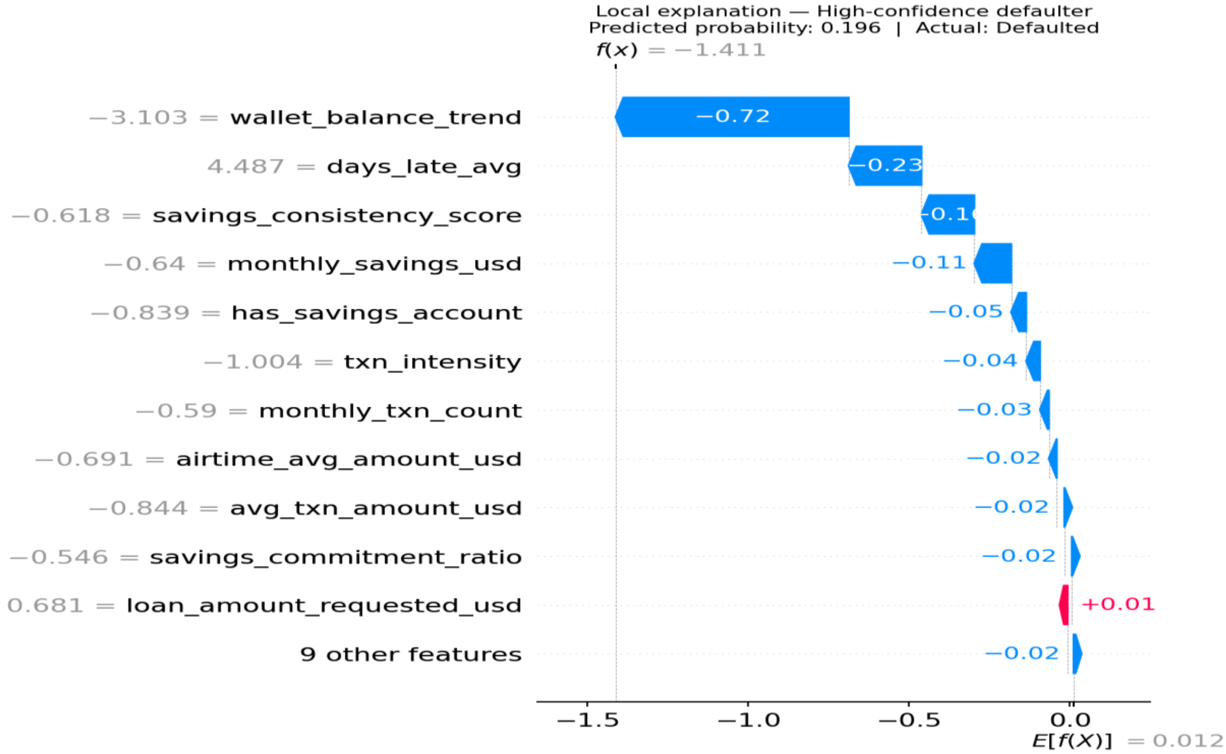

Figure 10b. Local SHAP explanation high-confidence defaulter. Red bars indicate features increasing default risk below the base rate.

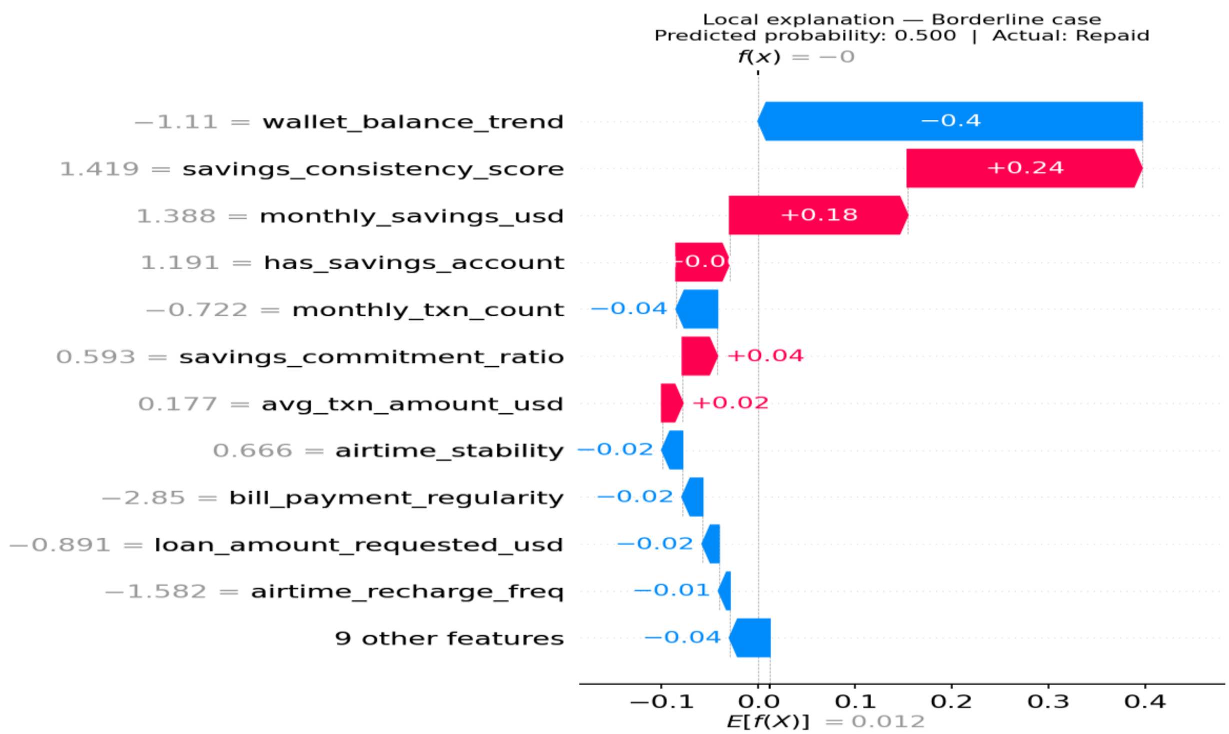

Figure 10c. Local SHAP explanation borderline borrower near the decision threshold. Competing positive and negative contributions illustrate model uncertainty at the margin.

This individual-level transparency is particularly important in the African lending context, where borrowers may be unfamiliar with algorithmic assessment and where regulatory frameworks increasingly require explainable credit decisions.

8. Fairness Analysis

8.1 Fairness criteria

We evaluate three criteria from the algorithmic fairness literature:

Demographic parity (Dwork et al., 2012): equal positive prediction rates across groups, assessed via selection rate disparity ratios.

Equal opportunity (Hardt et al., 2016): equal true positive rates across groups, ensuring that creditworthy borrowers have equal probability of approval regardless of demographic membership.

Predictive parity: equal precision across groups, ensuring that approved borrowers in each group have equal likelihood of actually repaying.

We apply the 80 percent rule as a disparity threshold: a disparity ratio below 0.80 is considered a meaningful fairness violation. In the absence of codified algorithmic parity thresholds within current Sub-Saharan African financial regulations, we adopt the United States Equal Credit Opportunity Act's 80 percent rule strictly as an international baseline indicator for methodological demonstration purposes. We acknowledge that no equivalent threshold has been formally adopted by regional bodies such as the Central Bank of Nigeria or the Central Bank of Kenya. However, recent compliance interventions such as the Central Bank of Kenya (Digital Credit Providers) Regulations 2022 signal an escalating regulatory focus on consumer privacy, algorithmic transparency, and predatory automated underwriting.

8.2 Results

Table 5 presents disparity ratios for regional and gender subgroups.

Table 5: Fairness disparity ratios by demographic group

Group	Selection rate	TPR	Precision
West Africa	1.000	1.000	0.990
East Africa	1.000	1.000	0.959
Southern Africa	1.000	1.000	1.000
Central Africa	1.000	1.000	0.989
Female	1.000	1.000	1.000
Male	1.000	1.000	0.977
Prefer not to say	1.000	1.000	0.989

All disparity ratios exceed 0.80 across all groups and all criteria. These properties are visualized in Figure 11 (Demographic Parity) and Figure 12 (Equal Opportunity), while Figure 13 confirms equitable predictive consistency via group-wise ROC-AUC. Demographic parity and equal opportunity are satisfied at 1.000

for every group, indicating that the model approves loans and identifies creditworthy borrowers at equal rates regardless of regional origin or gender.

We note that the "Prefer not to say" gender category warrants particular caution in real deployment contexts. Unlike the Male and Female categories, this category conflates actual gender identity with disclosure behavior individuals who decline to disclose their gender may differ systematically from those who do, meaning the category may itself encode behavioral or socioeconomic information (e.g., Zhang and Long, 2021). In a real lending system, this category should be treated as a distinct analytical group requiring separate study rather than as a third gender category. In the synthetic dataset, this category was generated independently of behavioral features and therefore does not exhibit this confounding.

To assess whether the fairness properties are robust to the choice of operating threshold, we evaluate disparity ratios at three additional approval rates spanning realistic microfinance lending contexts. Table 6 presents selection rate disparity ratios across regional and gender subgroups at four operating points.

Table 6: Selection rate disparity ratios across operating thresholds

Threshold	Approval rate	Region disparity	Gender disparity	Worst group
0.621	30%	0.818	0.932	West Africa
0.509	50%	0.811	0.947	Central Africa
0.433	70%	0.903	0.978	Central Africa
0.151	100%	1.000	1.000	—

All disparity ratios exceed the 0.80 threshold across every operating point evaluated, indicating that the fairness properties of the framework are robust to threshold selection and are not an artifact of the near-universal approval rate at the optimal threshold. The most constrained operating point 30% approval produces the lowest observed disparity ratio of 0.818 (West Africa, regional), which passes the 80% rule benchmark by a margin of only 0.018. This narrow margin warrants explicit caution: real-world mobile money data from West Africa may exhibit structural correlations between region and behavioral features due to infrastructure inequality, historical lending patterns, and urbanization differences. Such correlations would likely reduce this ratio below the 0.80 threshold, constituting a fairness violation under the same benchmark. This robustness analysis addresses the concern that single-threshold fairness evaluation may produce misleading results when the chosen threshold produces extreme selection rates (Corbett-Davies et al., 2017).

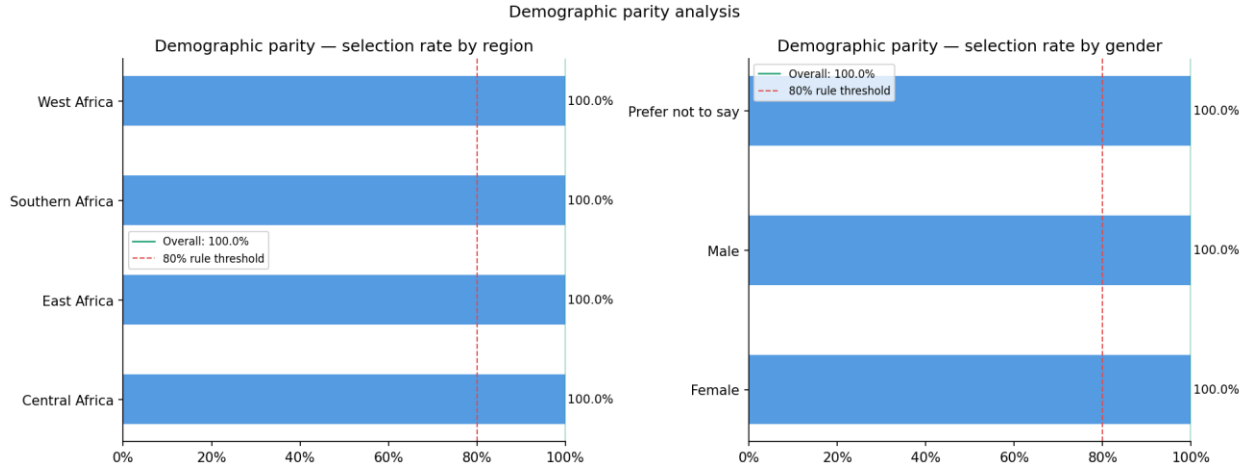

Figure 11. Demographic parity analysis. Selection rates by region (left) and gender (right). All groups exceed the 80% rule threshold (red dashed line), indicating equal approval rates across subgroups.

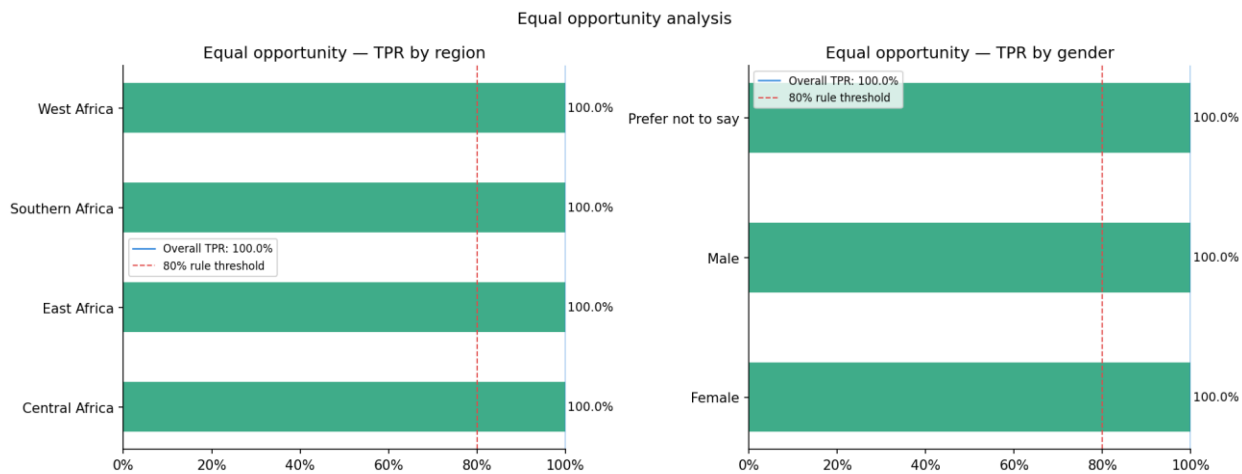

Figure 12. Equal opportunity analysis. True positive rates by region (left) and gender (right). All groups satisfy equal opportunity, confirming that creditworthy borrowers are identified at equal rates regardless of demographic membership.

8.3 Proxy analysis

Pearson correlations between the five highest-impact behavioral features and demographic group encodings were computed to assess whether behavioral features inadvertently encode demographic information. The maximum absolute correlation observed was 0.054 across all feature-group pairs, indicating no meaningful proxy relationships. This finding partially addresses the concern raised by Fuster et al. (2022) that behavioral proxies may perpetuate demographic discrimination through indirect pathways.

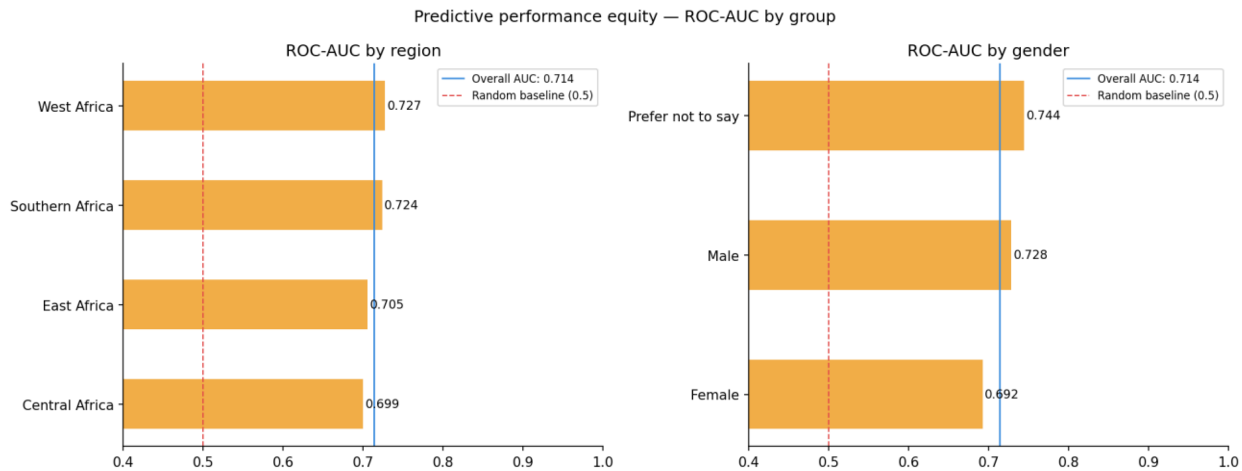

Figure 13. ROC-AUC by demographic group. All groups exceed the random baseline (0.5) and demonstrate consistent discrimination ability, indicating equitable predictive performance across subgroups.

8.4 Fairness limitations

These results must be interpreted with care. The synthetic data generating process explicitly constructed demographic attributes to be uncorrelated with behavioral features a condition that may not hold in real-world data, where structural inequality produces genuine correlations between geography, gender, and financial behavior. In a real deployment, fairness auditing on actual borrower data would be required before drawing conclusions about equitable treatment. Furthermore, we evaluate only three fairness criteria; satisfying all simultaneously is mathematically impossible when base rates differ across groups (Chouldechova, 2017), and the selection of criteria should ultimately reflect the values and legal context of the deploying institution. Contemporary fairness literature further cautions that optimizing for static group parity criteria can cause long-term economic harm by driving vulnerable populations into cycles of default, highlighting the need for dynamic or causal fairness formulations in future iterations (Corbett-Davies et al., 2017).

9. Discussion

9.1 Practical implications

Within the scope of the synthetic data generating process, the results demonstrate that behavioral financial data carries sufficient predictive signal to support the proposed framework's credit assessment pipeline. A ROC-AUC of 0.714 represents meaningful improvement over the random baseline (0.500), though the near-identical performance of logistic regression (0.713) and the accuracy below the majority-class baseline (0.755 vs 0.620) indicate that the threshold setting and model complexity require careful reconsideration before any deployment context. For a microfinance institution serving unscored populations, the practical value of the system lies not in replacing human judgment but in providing structured, explainable behavioral evidence to supplement it a role for which the SHAP explanation layer is particularly well suited.

The SHAP explanation layer addresses a critical gap in ML-based credit systems: the inability to communicate the basis of a decision to the borrower or loan officer. The waterfall and bar chart visualizations implemented in the dashboard translate statistical outputs into actionable language identifying, for example, that a borrower's declining wallet balance trend is the primary factor reducing their credit probability, which the borrower can act upon.

9.2 Limitations

Several limitations bound the conclusions of this study. First, the dataset is synthetic. While the data generating process is calibrated to reflect behavioral patterns documented in the literature, it cannot capture the full complexity of real mobile money data, including temporal dynamics, seasonal patterns, and network effects. Validation on real transaction data from a partner institution remains necessary.

Second, the feature set is constructed from literature-informed assumptions rather than empirical feature selection from real behavioral data. The relative importance of features may differ substantially across populations, geographies, and mobile money platforms.

Third, the fairness analysis is constrained by the synthetic data's designed independence between demographics and behavior. Real-world fairness properties cannot be inferred from this analysis.

Fourth, the system has not been evaluated for temporal stability. Credit scoring models are known to exhibit concept drift as economic conditions change, and a production system would require ongoing monitoring and retraining.

Fifth, the near-identical performance of logistic regression (ROC-AUC: 0.713) and tuned XGBoost (ROC-AUC: 0.714) suggests that the behavioral features in the synthetic dataset exhibit primarily linear structure. This limits the external validity of the SHAP non-linearity analysis in Section 7.3 and raises the question of whether XGBoost's additional complexity is justified. On real mobile money data with genuine non-linear feature interactions, the performance gap between linear and non-linear models may be larger.

Sixth, the engineered composite features (transaction intensity, savings commitment ratio, airtime stability) provided no measurable improvement over the 16 raw features in the ablation study (ΔAUC = -0.0002). While these features have sound theoretical motivation, their practical value on this synthetic dataset is negligible. Validation on real data is required to determine whether the engineering decisions add value in practice.

Seventh, the paper assumes that SHAP explanations reliably represent model behavior. Recent work has demonstrated that post-hoc explanation methods including SHAP can be manipulated to conceal systematic bias while producing plausible-looking explanations (Slack et al., 2020). This adversarial vulnerability means SHAP explanations should be treated as decision-support tools subject to auditing rather than as definitive proofs of model behavior.

Eighth, while TreeExplainer provides exact Shapley values relative to the tree ensemble's output, it assumes feature independence when computing conditional expectations. In the presence of correlated features — such as the savings-related features identified in Figure 3 SHAP values may distribute credit erroneously across correlated predictors, potentially misrepresenting individual feature contributions (Lundberg et al., 2020; Kumar et al., 2020).

9.3 Future work

Several extensions would strengthen this research. First, partnership with a microfinance institution or mobile money operator to validate the framework on real transaction data would address the most significant limitation. Second, longitudinal analysis incorporating time-series features from transaction histories could substantially improve predictive performance. Third, comparison with causal inference approaches to credit scoring addressing the confounding between behavioral features and unobserved creditworthiness represents a theoretically important extension. Fourth, the fairness framework could be extended to incorporate counterfactual fairness (Kusner et al., 2017), which provides stronger guarantees than the statistical parity criteria evaluated here. Fifth, implementing conditional expectation SHAP frameworks to mitigate feature dependency errors would improve individual explanation reliability on correlated feature sets. Sixth, incorporating adversarial robustness evaluation for SHAP explanations to detect potential manipulation of post-hoc explanations (Slack et al., 2020) and developing explanation methods resistant to such attacks represents an important direction for ensuring trustworthy credit decisions in high-stakes deployment contexts.

10. Conclusion

This paper presented FairLend-Africa, an explainable machine learning framework for alternative credit scoring using behavioral financial data. The system combines XGBoost-based prediction, SHAP-based individual explanation, and systematic fairness auditing in a deployable architecture accessible to resource-constrained institutions. On a synthetically generated dataset reflecting African mobile money behavioral patterns, the framework achieves a ROC-AUC of 0.7137, satisfies demographic parity and equal opportunity criteria across all evaluated subgroups, and provides individual-level explanations identifying wallet balance trend and savings consistency as the dominant creditworthiness signals under the synthetic data generating process. Comparison with a logistic regression baseline and feature ablation study reveal that the behavioral features exhibit primarily linear structure on synthetic data, motivating validation on real mobile money transaction data where non-linear interactions and genuine demographic correlations are expected to emerge.

The contribution of this work is methodological rather than empirical: we demonstrate that the combination of alternative behavioral data, gradient boosted trees, post-hoc explainability, and structured fairness auditing can be implemented as a coherent, reproducible, and deployable system within the resource constraints of independent research. Validation on real mobile money data from African lending institutions represents the critical next step toward practical application.

The complete codebase, dataset generation scripts, trained model artifacts, and experimental notebooks are available at: *https://github.com/highfrezh/fairlend-africa*

Appendix

Table A1: Optimized XGBoost hyperparameters

Hyperparameter	Value
learning_rate	0.05
max_depth	4
n_estimators	250
subsample	0.8
colsample_bytree	0.7
min_child_weight	5
scale_pos_weight	3.1

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